

Information Technology Institute (ITI)

Ministry of Communications and Information Technology — Egypt

Backend Node.js

Track Assignments

Generative AI & Prompt Engineering

Practical Assignments

Summer Training 2026

Confidential — For ITI Students Only

Task: Track-Specific AI Frameworks & Tooling

Scenario

The Generative AI ecosystem has evolved rapidly, introducing specialized frameworks and libraries to streamline the construction of RAG pipelines and Vector Database integrations within different tech stacks.

Requirements

- **Framework Comparison:** Research the two leading frameworks for building RAG pipelines in Node.js/TypeScript: **LangChain.js** and **LlamaIndex.TS**.
- **Evaluation:** Compare them based on ease of integration, flexibility with Vector Databases (Pinecone, ChromaDB, Qdrant), and built-in support for Chat History / Memory management.
- **Cost Optimization:** Research the concept of **Semantic Caching** in backend APIs (using Redis Vector Search) and explain how it reduces LLM API costs and query latency for repetitive requests.

Expected Deliverables

- **Documentation:** Provide official documentation PDF or Word.